

TEMPERATURE MONITORING & LCD DISPLAY

Arduino UNO + LM35 + 16×2 I2C LCD

1. Project Objective

To measure real-time temperature using an LM35 temperature sensor, display the current temperature on a 16×2 I2C LCD, and log the readings through the Arduino Serial Monitor.

2. Hardware Requirements

Component	Quantity
Arduino UNO	1
LM35 temperature sensor	1
16×2 I2C LCD	1
Breadboard	1
Jumper wires	As required
USB cable	1

3. Software Requirements

- Arduino IDE
- LiquidCrystal_I2C library
- USB serial monitor at 9600 baud

4. Wiring Connections

Component	Pin	Arduino UNO
LM35	VCC	5V
LM35	OUT	A0
LM35	GND	GND
I2C LCD	VCC	5V
I2C LCD	GND	GND
I2C LCD	SDA	A4
I2C LCD	SCL	A5

LM35 orientation: With the flat face toward you, the commonly used TO-92 LM35 has VCC on the left, OUT in the middle, and GND on the right. Check your specific sensor's datasheet before powering it.

5. Arduino Program

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

LiquidCrystal_I2C lcd(0x27, 16, 2);
const int tempPin = A0;

void setup()
{
    Serial.begin(9600);

    lcd.init();
    lcd.backlight();

    lcd.setCursor(0, 0);
    lcd.print("Temperature");
    lcd.setCursor(0, 1);
    lcd.print("Monitoring...");
    delay(2000);
    lcd.clear();
}

void loop()
{
    int sensorValue = analogRead(tempPin);

    // Convert ADC value to voltage
    float voltage = sensorValue * (5.0 / 1023.0);

    // LM35 gives 10 mV per degree Celsius
    float temperature = voltage * 100.0;

    // Display on LCD
    lcd.setCursor(0, 0);
    lcd.print("Temperature:");

    lcd.setCursor(0, 1);
    lcd.print(temperature, 1);
    lcd.print((char)223);
    lcd.print("C    ");

    // Display on Serial Monitor
    Serial.print("Temperature: ");
    Serial.print(temperature, 1);
    Serial.println(" C");

    delay(1000);
}
```

6. Program Explanation

#include — Enables I2C communication used by the LCD.

#include — Provides functions to control the I2C LCD.

LiquidCrystal_I2C lcd(0x27, 16, 2); — Creates a 16×2 LCD object at the common I2C address 0x27.

analogRead(A0) — Reads the analog voltage produced by the LM35.

voltage = sensorValue × 5.0 / 1023.0 — Converts the 10-bit ADC value into voltage, assuming a 5 V ADC reference.

temperature = voltage × 100 — Converts LM35 voltage to °C because its nominal sensitivity is 10 mV/°C.

Serial.print() — Sends the measured temperature to the Serial Monitor.

delay(1000) — Updates the measurement approximately once every second.

7. Working Principle

The LM35 senses the surrounding temperature and produces an analog voltage proportional to temperature. Arduino UNO reads this voltage through analog input A0. The ADC converts the voltage into a digital value. The program then calculates temperature in degrees Celsius and sends the result to both the I2C LCD and Serial Monitor.

Flow: LM35 → A0 of Arduino → Temperature calculation → 16×2 I2C LCD + Serial Monitor

8. Expected LCD Output

```
Temperature:
25.4°C
```

The displayed value will change according to the measured temperature.

9. Expected Serial Monitor Output

```
Temperature: 25.4 C
Temperature: 25.5 C
Temperature: 25.5 C
Temperature: 25.6 C
```

10. Deliverables Checklist

- Working Arduino circuit
- Arduino sketch uploaded successfully
- 30–60 second video showing live LCD reading
- One-page README containing wiring and explanation
- Serial Monitor showing temperature readings

11. Applications

- Room temperature monitoring
- Environmental monitoring
- Smart home systems
- Electronic equipment temperature monitoring
- Basic IoT monitoring systems

12. Result

The temperature monitoring system successfully measures temperature using the LM35 sensor and displays the real-time value on a 16×2 I2C LCD. The same readings are also logged through the Arduino Serial Monitor.

Note

If the LCD does not display anything, the I2C address may differ from 0x27 (commonly 0x3F on some modules). An I2C scanner can be used to find the correct address.